01 - Introduction

Pattern Recognition and Machine Learning, Lab 1

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What you will learn

Part 1

Decision making

- Bayesian decision theory
- Non-Bayesian tasks

Probabilities are **given**

Part 2

Parameter estimation

- MLE and MAP estimation
- Parzen windows, k-NN

Probabilities are **estimated**

Part 3

Classifiers and learning

- Logistic regression
- Perceptron, SVM
- AdaBoost, neural nets
- K-means, EM, PCA
- Decision trees

Many **different principles**

There is a lot. Do not memorize it, understand how the methods compare and relate.

Organization

Lab overview

- **Weekly assignment**: implement a method from the lecture
- Labs are mostly **exercises** at the whiteboard
- **Three tests** during the semester, bring a calculator!
- It is a **difficult class**!

Lab plan 2026/27

25.9.	Introduction, Python, NumPy	
2.10.	Bayesian decision task	
9.10.	Non-Bayesian tasks: minimax	
16.10.	**Parzen windows**	**test**
23.10.	MLE, MAP, Bayesian estimation	
30.10.	Logistic regression	
6.11.	Problem solving	
13.11.	**Linear classifier: perceptron**	**test**
20.11.	*Dean's day, no lab*	
27.11.	Support Vector Machine	
4.12.	AdaBoost	
11.12.	**K-means clustering**	**test**
18.12.	Convolutional neural networks	
8.1.	Credits, revision, problem solving	

Expectations

Time

- **6 credits ≈ 156 hours** in the semester
- About 1 hour/week on the lecture
- 11 lab assignments ≈ **6-7 hours/week**

Plan enough time for the assignments!

Skills

- **Python**: NumPy, Jupyter notebooks
- **Mathematics**
  - probability and statistics
  - derivatives, integrals
  - linear algebra, optimization

Credit requirements (*zápočet*)

All 11 assignments

Each one must **pass the BRUTE evaluation**

Final deadline: **10 January 2027**

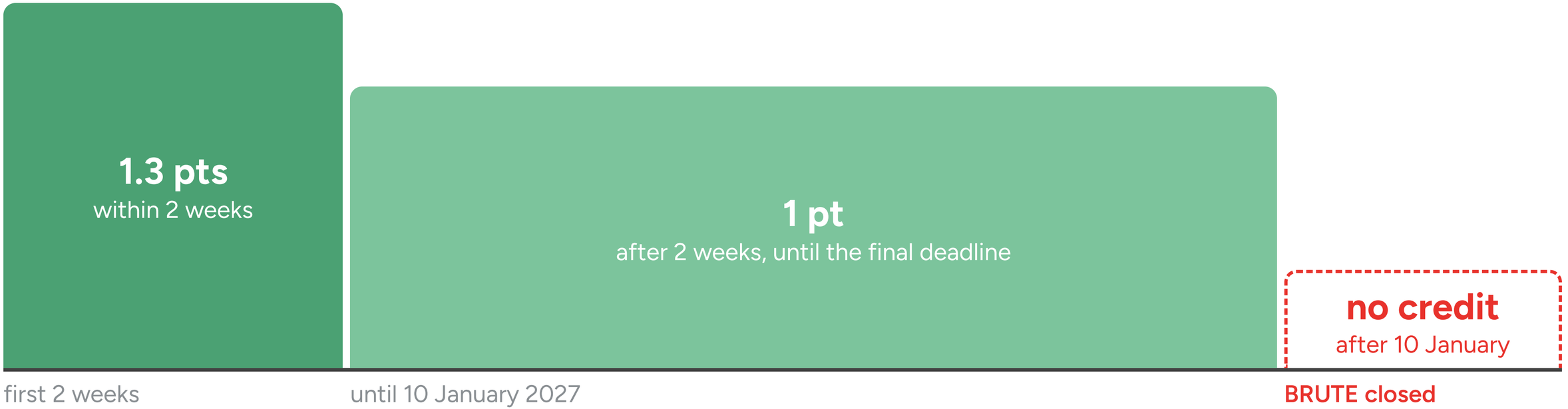
All 3 tests

Attend the short written tests at the beginning of the labs on

16 October, 13 November and 11 December

Missing a test for a serious reason? Contact your TA to arrange a make-up test.

Points for an assignment



The introductory assignment (this week) is worth **1 point**, so 1 + 10 × 1.3 = **14 points** in total.

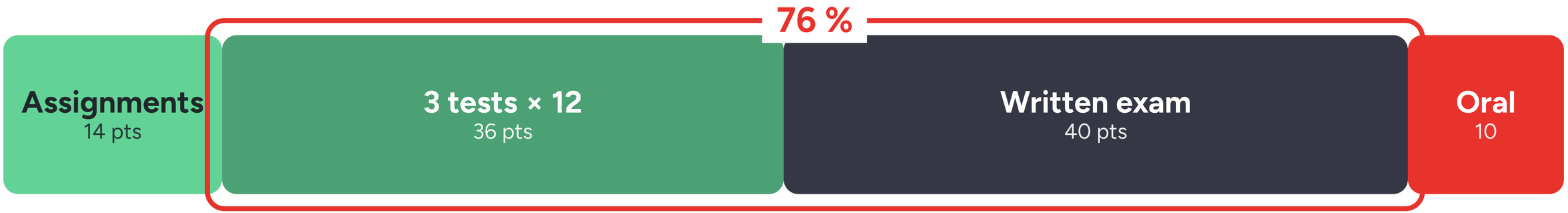
Bonus tasks: up to **+0.8 pts** each. Show them at the lab, no deadline (until the day before the exam).

Final grade: 100 points



- Points from the labs go **directly** into the final evaluation
- The **oral exam is compulsory**
- You can take the exam only **with the zápočet**

Final grade: 100 points



+ bonus Bonus points can push the lab score **above 50** or compensate for points lost in the tests

76 % of the grade comes from written work: tests + written exam, i.e. "math" and "problem solving"

Grade scale

F	E	D	C	B	A
< 50	50-59	60-69	70-79	80-89	90+

- A **minimum number of points** from the written exam is required to pass
- The oral exam can move your grade in **both directions**
- Exam questions from previous years and example oral questions are on the course page (Czech, English) and in the exam question archive

Support

CourseWare

- Lab instructions
- **Git templates**
- **BRUTE** upload system

Forum

The preferred channel, **all TAs answer there**.

- do not share your code
- do not share derivations
- search first

E-mail your TA

Only **personal** matters:

- missed test, illness
- no forum access

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Never e-mail your code asking "what is wrong?". Ask at the forum, TAs see your code.

Cooperation and plagiarism

OK

- Study together, use the **Exercise book** (problems from previous tests)
- Discuss lectures, exercises and labs
- Discuss implementation ideas

Not OK

- Using a piece of implementation or derivation found anywhere
  - except course materials and tutorials
- Sharing or publishing your solution

Plagiarism = no zápočet

AI tools

- AI tools are **generally allowed**, unless stated otherwise
- **You are the author**, including the mistakes AI makes
- Understand **every line**, we may ask you to explain it
- Copying longer AI-generated parts counts as **plagiarism**
- Be open about how you used AI

Best practice

Use AI to learn the syntax or an idea, then **implement the solution again from your head** (e.g. after 24 hours).

Our goal

We want you to learn, not to paste the assignment into a chat.

Avoid *vibe-coding*.

Lab 1: assignment

- ☑ **Install** your IDE (PyCharm, VS Code, Vim, ...)
- ☑ Verify access to the **forum**, **git templates** and **BRUTE**
- ☑ Download the **git template** for the first lab
- ☑ Verify that you can **upload to BRUTE** and get the results back
- ☑ Get familiar with **Python** and **NumPy**

Do NOT forget: there is homework already from this first lab!

Recap from probability and statistics

Probability

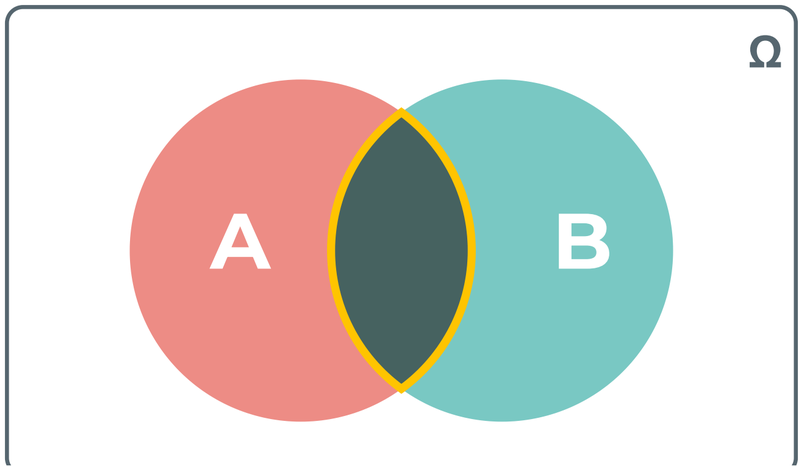
A1: Non-negativity $P(A) \geq 0$

A2: Normalization $P(\Omega) = 1$

A3: Additivity $P(A \cup B) = P(A) + P(B) \text{ if } A \cap B = \emptyset$

In general: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$P(A \cap B) = ?$



Two pens, pigs and sheep

P() = 5/10 = ½

P() = 5/10 = ½

P([ ]) = 4/10 = ⅖

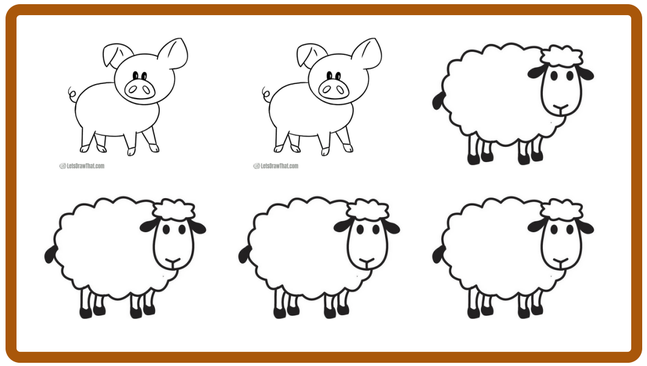
P([ ]) = 6/10 = ⅗

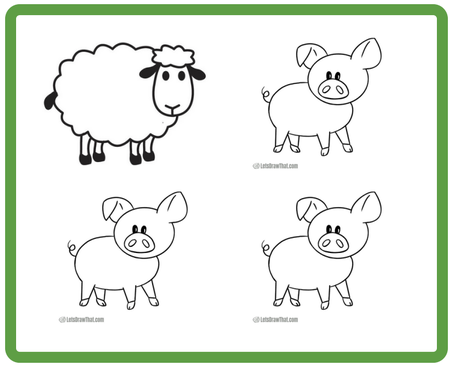
P( | [ ]) = 3/4

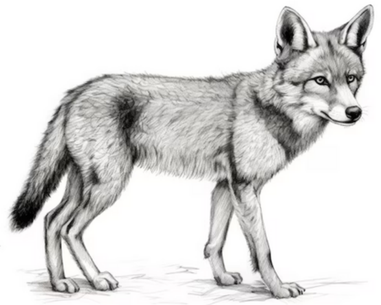
P( | [ ]) = 4/6 = ⅔

P([ ] | ) = 3/5

P([ ] | ) = 4/5







The wolf is hungry. He picks a random animal for lunch.

? The wolf ate a **sheep**. What is the probability that it came from the **brown** pen?

Total probability

Product rule

$$P(A \cap B) = P(A, B) = P(A \mid B)\, P(B) = P(B \mid A)\, P(A)$$

Sum rule, also called **marginalization**

$$P(A) = P(A, B) + P(A, B^c)$$

B^c is the **complement** of B: all outcomes in which B does not occur, $P(B^c) = 1 - P(B)$

Total probability

$$P(A) = \sum_i P(A, B_i) = \sum_i P(A \mid B_i)\, P(B_i)$$

B_i are mutually exclusive and jointly exhaustive (exactly one of them occurs)

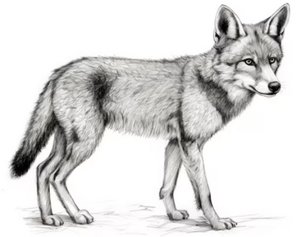
More pigs and sheep

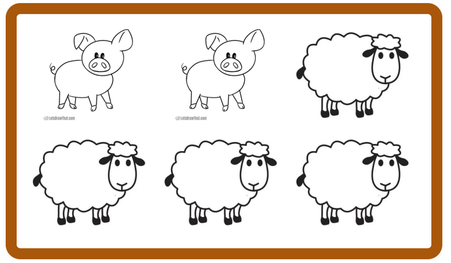
P(🐖, [ ]) = P(🐖 | [ ]) P([ ]) = ¾ · ⅖ = **0.3**

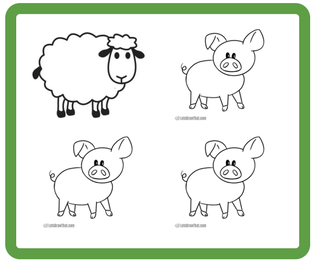
P([ ] | 🐖) P(🐖) = ⅗ · ½ = **0.3**

P(🐖, [ ]) = P(🐖 | [ ]) P([ ]) = 2/6 · ⅗ = **0.2**

P(🐖) = P(🐖, [ ]) + P(🐖, [ ]) = 0.3 + 0.2 = **0.5**







? We know both joint probabilities. What is the probability that the wolf eats a **pig** at all?

Bayes theorem

$$P(H \mid E) = \frac{P(E \mid H)\, P(H)}{P(E)}$$

H: hypothesis, e.g. it will rain today
E: evidence, e.g. the morning is cloudy

posterior
$P(H \mid E)$ belief in H after seeing the evidence

rain, given a cloudy morning

prior
$P(H)$ belief in H before any evidence

how often it rains

likelihood
$P(E \mid H)$ how well H explains the evidence, *not* a probability of H!

cloudy morning, given rain

marginal
$P(E)$ total probability of the evidence, often hard to compute

how often it is cloudy

Later in RPZ: θ = model parameters, D = data:

$$P(\theta \mid D) = P(D \mid \theta)\, P(\theta) \,/\, P(D) \;\propto\; P(D \mid \theta)\, P(\theta)$$

In words: the posterior of the parameters given the data is proportional to the likelihood of the data times the prior of the parameters. $P(D)$ does not depend on θ, so we can drop it when comparing parameters.

Example: mammogram

In a study, physicians were asked about a woman with an initial **1 % risk of cancer** (a malignant tumor) whose **mammogram came back positive**.

A mammogram correctly classifies about **80 % of malignant** tumors (positive result) and **90 % of benign** tumors (negative result).

95 of 100 physicians estimated the probability of cancer at about **75 %**.

Do you agree?

Source: M. Merling, *Practice questions on Bayes's formula*, UPenn Math 107

Example: mammogram, solution

M: malignant, $B = M^c$: benign

$+$: positive result

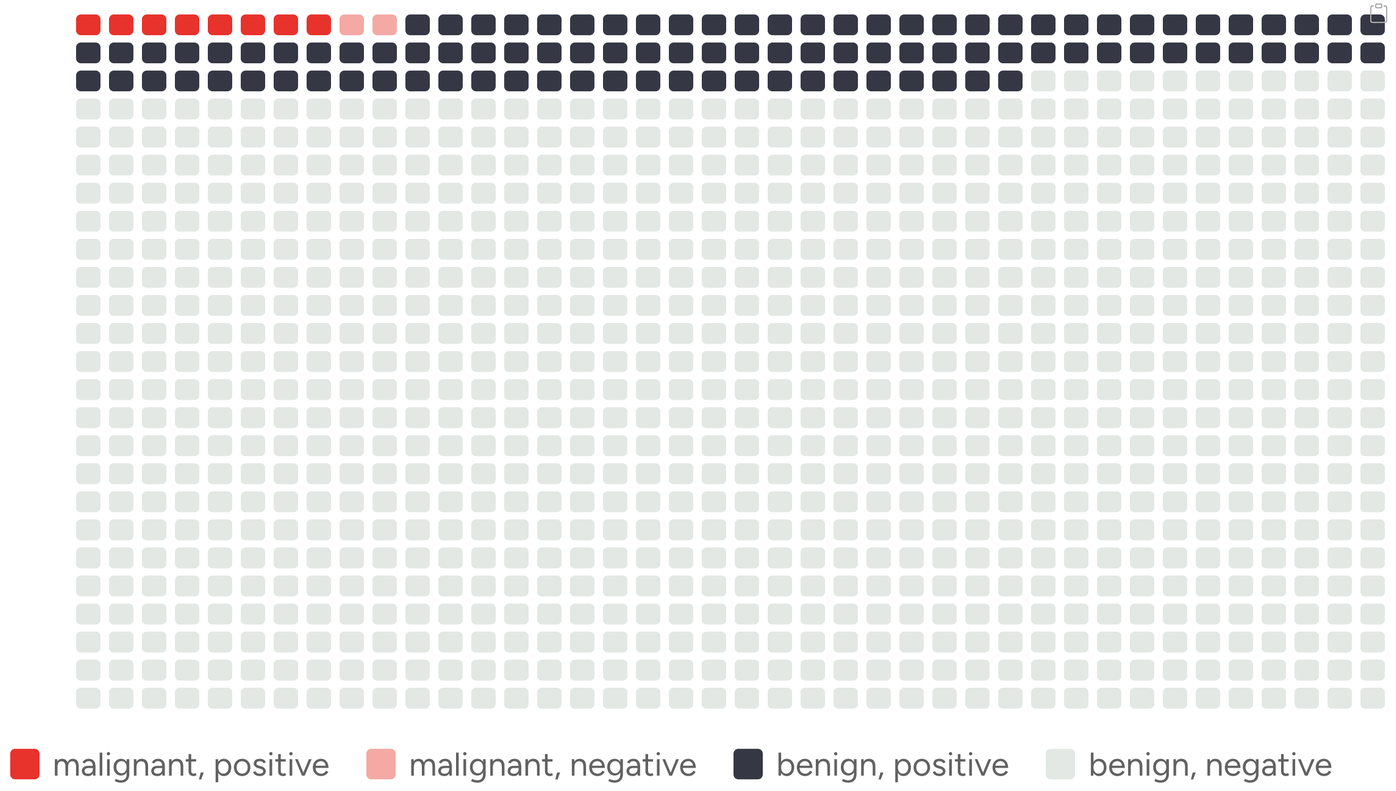
Known: $P(M) = 0.01$, $P(+ \mid M) = 0.8$, $P(- \mid B) = 0.9$

$\Rightarrow P(+ \mid B) = 0.1$

P(M)	0.01
P(+ | M)	0.8
P(– | B)	0.9

P(M | +) = **7.5 %**

Of 1000 women: **8** true vs. **99** false positives.



$$P(M \mid +) = \frac{P(+ \mid M)\,P(M)}{P(+ \mid M)\,P(M) + P(+ \mid B)\,P(B)} = \frac{0.8 \cdot 0.01}{0.8 \cdot 0.01 + 0.1 \cdot 0.99} \approx 0.075$$, the prior matters!

Example: three cards

We have **3 cards** that are identical in form, except for their colors:



RR: both sides red

BB: both sides black

RB: one side red, one black

The cards are mixed in a hat, one is drawn at random and put on the ground.

If the **upper** side of the chosen card is **red**, what is the probability that the **other** side is **black**?

Source: M. Merling, *Practice questions on Bayes's formula*, UPenn Math 107

Example: three cards, solution



RR BB RB

All **6 sides** are equally likely to be on top.

We see red, so only the **3 red sides** remain possible.

Only **one** of them has black on the other side: $P = \frac{1}{3}$, not $\frac{1}{2}$!

Formally, with R = "the upper side is red":

$$P(RB \mid R) = \frac{P(R \mid RB)\,P(RB)}{P(R \mid RR)\,P(RR) + P(R \mid RB)\,P(RB) + P(R \mid BB)\,P(BB)} = \frac{\frac{1}{2} \cdot \frac{1}{3}}{1 \cdot \frac{1}{3} + \frac{1}{2} \cdot \frac{1}{3} + 0 \cdot \frac{1}{3}} = \frac{1}{3}$$

Extra: videos

Bayes theorem, the geometry of changing beliefs 3Blue1Brown

The quick proof of Bayes' theorem 3Blue1Brown

The medical test paradox, and redesigning Bayes' rule 3Blue1Brown

The Bayesian Trap Veritasium

Conditional Probabilities, Clearly Explained StatQuest

Bayes' Theorem, Clearly Explained StatQuest



3Blue1Brown

More from 3Blue1Brown: 3blue1brown.com/topics/probability


"That's all Folks!"